THE GAUDIUM SCHOOL
Sculpting happy minds

Students Name:	Grade & Sec: MYP-1	Date: -08-2026
Subject: Integrated Sciences	Task Duration: 1 hour	Level Obtained
Unit Title :		

Criterion-A—Knowing and understanding

i. **outline** scientific knowledge
ii. **apply** scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations
iii. **interpret** information to make scientifically supported judgments

Instructions

- Answer all questions.
- Use scientific vocabulary where appropriate.
- For questions that ask you to explain, analyse, evaluate, or justify, support your answers using particle theory.
- Show your scientific reasoning clearly.

CRITERION-A : Knowing and Understanding

Achievement Level	Level Descriptor	Task-Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard
1-2	**The student should be able to:** i. select scientific knowledge ii. select scientific knowledge and understanding to suggest solutions to problems set in familiar situations iii. apply information to make judgments, with limited success	**The student is able to:** **select** scientific knowledge about states of matter and their particle arrangement **select** scientific knowledge and understanding to suggest a simple solution to a familiar problem involving melting **apply** information to make a judgment, with limited success, about physical and chemical

Cambridge Assessment International Education
Cambridge International School

		changes.
3-4	**The student should be able to:** i. recall scientific knowledge ii. apply scientific knowledge and understanding to suggest solutions to problems set in familiar situations iii. apply information to make judgments	**The student is able to:** **recall** scientific knowledge about the arrangement, spacing and movement of particles **apply** scientific knowledge about freezing to suggest a solution to a familiar problem **apply information to make judgments** about how thermal energy affects changes of state.
5-6	**The student should be able to:** i. state scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in familiar situations iii. apply information to make scientifically supported judgments	**The student is able to:** **state** scientific knowledge about physical and chemical changes using appropriate scientific reasoning **apply** particle theory to solve familiar problems involving melting **apply information to make scientifically supported judgments** about factors affecting evaporation using evidence and particle theory.
7-8	**The student should be able to:** i. outline scientific knowledge ii. apply scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations iii. interpret information to make scientifically supported judgments	**The student is able to:** **outline** scientific knowledge about changes of state using particle movement, arrangement and energy **apply** scientific knowledge and understanding to solve problems and **suggest scientifically appropriate solutions to unfamiliar situations** involving changes of state **interpret information** to make scientifically supported judgments about factors affecting melting using evidence and particle theory, and determine whether the evidence is

		sufficient to support a conclusion.

CRITERION-A : Knowing and Understanding

Q1 – The diagrams below show three substances.

- **A:** particles are closely packed and arranged regularly.
- **B:** particles are close together but can move past one another.
- **C:** particles are far apart and move freely.

Select the substance that represents a **liquid**.

A. Substance A
B. Substance B
C. Substance C

Strand i | Achievement Level 1–2

____

____

____

Q2 Recall your scientific knowledge about liquids. Describe the arrangement, spacing and movement of particles in a liquid. **Strand i | Achievement Level 3–4**

____

____

____

Q3 State why gases can be compressed easily but liquids cannot be compressed easily. Give a scientific reason using the spacing between particles.

Strand i | Achievement Level 5–6

____

____

Q4 Water vapour in the air changes directly into tiny ice crystals on the inside wall of a freezer. Using the particle model, outline what happens to the movement, arrangement and energy of the particles during this change. Name the change of state. **Strand i | Achievement Level 7–8**

Q5 A student wants to prevent an ice cube from melting quickly during a science activity.

She can place it:

Option A: on a metal tray near a sunny window
Option B: inside a closed insulated container

Select the better option and suggest why it would help the ice cube remain solid for longer.

Strand ii | Achievement Level 1–2

____

____

Q6 A family wants to freeze homemade juice into ice pops as quickly as possible.

They can place the mould:

A: near the freezer door, where the temperature changes frequently
B: deep inside the freezer, where the temperature remains very low

Apply your knowledge of freezing to suggest the better position. Give a scientific reason.

Strand ii | Achievement Level 3–4

____

____

____

____

____

Q7 A shopkeeper notices that frozen food begins to soften when the freezer door is left open for a long time.

Apply your knowledge of particle theory to explain why the frozen food begins to melt and solve the problem by suggesting one action the shopkeeper should take to reduce melting.

Strand ii | Achievement Level 5-6

____

____

____

____

____

Q8 On very cold nights, water vapour in the air can change directly into ice crystals and form frost on young plants. Frost can damage the plants.

A farmer notices that frost is expected overnight.

Apply your knowledge of changes of state to:

a) Identify the change of state that forms frost and **describe the change in particle movement, arrangement and energy**.

b) suggest one method the farmer could use to protect the plants from frost and explain scientifically how the method would help.

Strand ii | Achievement Level 7–8

____

____

____

____

____

Q9 A**A student observes three substances.**

Substance	**Observation**
A	Keeps its own shape
B	Takes the shape of its container but keeps the same volume

C	Spreads out to fill the entire container

A student says:

"Substance B is a gas."

Apply the information in the table to make a judgment about whether the statement is correct. Give one piece of evidence from the table.

Strand iii | Achievement Level 1–2

____

____

____

Q10 Two students leave equal amounts of ice in different places.

Cup	Location	Time taken for ice to melt
A	Sunny windowsill	12 minutes
B	Shaded table	28 minutes

A student concludes:

"Ice melts faster when it receives more thermal energy."

Apply the information in the table to make a judgment about the student's conclusion.

Give a scientific reason for your judgment.

Strand iii | Achievement Level 3–4

____

____

Q11 A student places equal amounts of water in three containers.

Container	Surface area	Water remaining after 3 hours
A	Small	45 mL
B	Medium	38 mL
C	Large	29 mL

All three containers started with 50 mL of water and were kept at the same temperature.

The student concludes:

"Increasing the surface area increases the rate of evaporation."

Apply the information to make a scientifically supported judgment about the student's conclusion.

In your answer:

- use two pieces of evidence from the table
- use particle theory to support your judgment.

Strand iii | Achievement Level 5–6

____

____

Q12 – A student investigates how temperature affects the melting of equal-sized ice cubes.

Trial	Temperature of surroundings	Size of ice cube	Time taken to melt
A	15°C	Equal	22 min
B	25°C	Equal	14 min
C	35°C	Equal	7 min

The student concludes:

"Increasing the temperature of the surroundings makes ice melt faster."

Interpret the information in the table to make a scientifically supported judgment about the student's conclusion.

In your answer:

1. Use at least two pieces of evidence from the table.
2. Use particle theory to explain the pattern.
3. Decide whether the evidence is sufficient to support the student's conclusion.

Strand iii | Achievement Level 7-8

____

____

____